Viviana Carcaiso

Email: viviana.carcaiso@inrae.fr
Webpage: vivianacarcaiso.github.io
Scholar: Google Scholar

Postdoctoral Researcher

INRAE – Unité Biostatistique et Processus Spatiaux, Avignon, France

Research Interests: Extreme value theory, Bayesian statistics, statistical learning for extremes, environmental and climate applications.

Academic Positions

2024–present Postdoctoral Researcher, BioSP unit, INRAE, France.

2024: Visiting Research Student, University of Edinburgh, UK (6 months).

2021–2025 Ph.D. Student, University of Padova, Italy.

Education

2021–2025 Ph.D. in Statistical Sciences, University of Padova
Thesis: Bayesian Mixture Models for Extremes
Supervisor: Ilaria Prosdocimi
Co-supervisors: Isadora Antoniano-Villalobos, Miguel de Carvalho

2019–2021 M.Sc. in Statistics and Data Science, University of Florence *(summa cum laude)*
Thesis: The impact of remote teaching on university students' gained credits: an analysis based on parametric modeling of quantile regression coefficient functions
Supervisor: Leonardo Grilli

2016–2019 B.Sc. in Statistics, University of Florence *(summa cum laude)*

Grants and Awards

2026 **ISBA** EnviBayes Student Paper Competition Winner.

2025–present Research assistant in project CLIMATHS, PEPR Maths-Vives, France 2030 investment plan.

2024–2025 Research assistant in a project funded by H2020 FIRE-RES.

2021 PhD Scholarship, University of Padova (3 years).

Research

Papers under Revision

Carcaiso, V., de Carvalho, M., Prosdocimi, I., Antoniano-Villalobos, I. Bayesian mixture models for heterogeneous extremes. *Journal of Agricultural, Biological and Environmental Statistics (major revision)*.

Articles in Journals

Richards, J., Lee, M. W., **Carcaiso, V.** & de Carvalho, M. (2025). Contribution to the Discussion of 'Inference for extreme spatial temperature events in a changing climate with application to Ireland' by Healy et al. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 74(2), 307–309.

Carcaiso, V., Grilli, L. (2022). Quantile regression for count data: jittering versus regression coefficients modelling in the analysis of credits earned by university students after remote teaching. *Statistical Methods & Applications*, 1–22.

Book Chapters

de Carvalho, M., **Carcaiso, V.** (2026). Learning about extreme value distributions from data. In de Carvalho, M., Huser, R., Naveau, P., and Reich, B. J. (Eds.), *Handbook on Statistics of Extremes*. Chapman & Hall/CRC, Boca Raton, FL.

Conference Proceedings

Carcaiso, V., Prosdocimi, I., Antoniano-Villalobos, I. (2023). Regression for mixture models for extremes. *Book of Short Papers, SIS 2023 – Statistical Learning, Sustainability and Impact Evaluation*, 629–634.

Carcaiso, V., Grilli, L. (2022). Quantile regression coefficient modeling for counts to evaluate the productivity of university students. *Book of Short Papers, SIS 2022*, 1333–1338.

Carcaiso, V., Grilli, L. (2022). Analysis of count data by quantile regression coefficient modelling: student's gained credits after online teaching. *Proceedings of the 36th International Workshop on Statistical Modelling*, 113–116.

Ongoing Research

Carcaiso, V., Engelke, S., Legrand, J., Opitz, T. Extrapolation of extreme covariates in generalized additive models using extreme-value theory. *(manuscript in preparation)*

Carcaiso, V., Bouchet, F., Fischer, A., Opitz, T., Yiou, P. Simulation and probabilistic analysis of complex extreme events with applications to compound climate extremes in agronomy and ecology.

Invited Talks

September 2025 RISE Final Workshop – *Bayesian mixture models for heterogeneous extremes*. Venezia Mestre, Italy.

June 2025 Extreme Value Analysis (EVA 2025) – *Extrapolation of extreme covariates with an application to wildfire prediction*. Chapel Hill, USA.

December 2024 IMS International Conference on Statistics and Data Science (ICSDS) – *Infinite mixture models for heterogeneous environmental extremes*. Nice, France.

December 2023 Computational and Methodological Statistics (CMStatistics) – *Where do extremes come from? Dependent mixtures for block maxima*. Berlin, Germany.

Conference Presentations

Contributed Talks

June 2026 – 57^es^ Journées de Statistique de la SFdS, Clermont-Ferrand, France.

September 2025 – GRASPA 2025, Rome, Italy.

July 2024 – Extreme Value Analysis and Application to Natural Hazards (EVAN), Venice, Italy.

September 2023 – STOR-i Extremes Workshop (STEW), Lancaster, UK.

June 2023 – Statistical Learning, Sustainability and Impact Evaluation (SIS 2023), Ancona, Italy.

July 2022 – International Workshop on Statistical Modelling (IWSM), Trieste, Italy.

Poster Presentations

December 2025 – Conference on Stochastic Weather Generators (SWGEN), Grenoble, France.

July 2024 – 2024 ISBA World Meeting, Venice, Italy.

June 2024 – Centre for Statistics Annual Conference, Edinburgh, UK.

June 2023 – Extreme Value Analysis (EVA 2023), Milan, Italy.

Teaching Experience

Instructor

Statistical Distribution Models for Extreme Events: Why and How to Use Them?
Workshop *Extreme Events in Agriculture and Forestry*, Avignon, France – June 2026

Developed course materials and delivered a short course on statistical models for extreme events, covering both theory and implementation in R for an interdisciplinary audience in agriculture and environmental sciences.

Research Visits

16–21 July 2025 Université de Bretagne Occidentale, Brest, France. Visiting Juliette Legrand.

13–16 May 2025 University of Geneva, Geneva, Switzerland. Visiting Sebastian Engelke.

January–June 2024 University of Edinburgh, Edinburgh, UK. Visiting Miguel de Carvalho.

Academic Activities

Memberships

Italian Statistical Society (SIS).

International Society for Bayesian Analysis (ISBA).

EnviBayes (ISBA Section on Bayesian Methods in Environmental and Ecological Sciences).

CIRCE (Community of Italian Researchers Connecting on Extreme Value Statistics).

Peer Review

Reviewer, *Stochastic Environmental Research and Risk Assessment* (SERRA).

Reviewer of two chapters, *Handbook on Statistics of Extremes*, Chapman & Hall/CRC.

Event Organization

Organizing committee, workshop *Extreme Events in Agriculture and Forestry*, Avignon, France (2026).

Volunteer, *ISBA World Meeting 2024*.

Technical Skills

Languages: Fluent in Italian (native) and English, intermediate in French.

Programming & Software: R and LaTeX (daily use); Python, SAS, Stata, Fortran, QGIS, LimeSurvey, Microsoft Office.